higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

DIGITAL ELECTRONICS N4

(8080354)

5 December 2024 (X-paper)
09:00–12:00

Drawing instruments may be used. Calculators may NOT be used.

This question paper consists of 7 pages.

254Q1E2405

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
DIGITAL ELECTRONICS N4
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a new page.
5. Use only a black or blue pen.
6. Write neatly and legibly.

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SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 $2,2_8 - 2_{16}$ =

A 0,5
B 0,25
C 0,75
D 1

1.2 $10,01_2 \times 4_{16}$ =

A 1001_2
B 1001_8
C 1001_{10}
D 1001_{16}

1.3 $100.001_2 + 4,1_{16}$ =

A $8,3_8$
B $10,14_8$
C $8,3_{16}$
D $10,14_{16}$

1.4 $11_8 \div 2,4_{16}$ =

A 100_2
B $100,11_2$
C $100,01_2$
D $11,11_2$

1.5 $20,2_8 \div 4,04_8$ =

A 1_8
B 2_8
C 4_8
D 8_{16}

(5 × 2) **[10]**

QUESTION 2

Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (2.1–2.8) in the ANSWER BOOK.

2.1 $\overline{A} + B\overline{B} + 0 = \overline{A}$

2.2 $\overline{A} + \overline{A}B = \overline{A} + B$

2.3 $A + \overline{B} + \overline{A} + C = 0$

2.4 $X + X + 0 = X$

2.5 $X \bullet X = X$

2.6 $A\overline{A} + AB + 0 = AB$

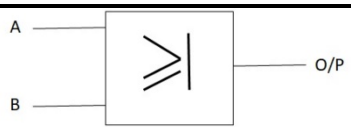
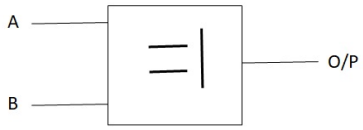
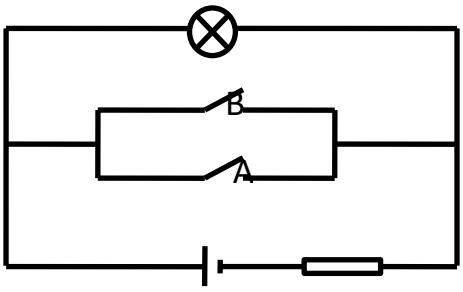
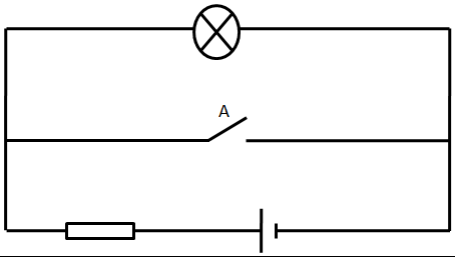
2.7 $X + X + 1 = X$

2.8 $\overline{A} + \overline{A}B = \overline{\overline{A} \bullet B}$

(8 × 1) **[8]**

QUESTION 3

Choose a/an term/item from COLUMN B that matches a description in COLUMN A. Write only the letter (A–G) next to the question number (3.1–3.7) in the ANSWER BOOK.

COLUMN A	COLUMN B
3.1 	A AND GATE
3.2 $\overline{A \bullet B} = O/P$	B OR GATE
3.3 	C EXCLUSIVE OR GATE
3.4	D NOT GATE
3.5 	E NAND GATE
3.6	F NOR GATE
3.7 	G EXCLUSIVE NOR GATE

A	B	O/P
0	0	0
0	1	0
1	0	0
1	1	1

A	B	O/P
0	0	1
0	1	0
1	0	0
1	1	1

(7 × 1) **[7]**

TOTAL SECTION A: 25

SECTION B

QUESTION 4

Complete the following calculations in the binary numbering system. All steps must be indicated clearly.

4.1 $11101{,}01111_2$ – $1110{,}1001_2$ (Make use of the 1s complement method.) (5)

4.2 $101{,}011_2$ – $11010{,}1001_2$ (Make use of the 2s complement method.) (5)

4.3 $100011{,}111_2$ × $110{,}101_2$ (5)

4.4 $10100101{,}1001111_2$ ÷ $101{,}011_2$ (5)

[20]

QUESTION 5

Draw the circuit diagrams of the Boolean expressions below. Do not change or simplify the expressions:

5.1 $\overline{\overline{(A \,.\, B)}.\,\overline{(B + \overline{C})} + \overline{A \oplus C}} = f$ (5)

5.2 $\overline{\overline{(A + \overline{B})}.\,\overline{(A + \overline{B} + \overline{C})}.\,\overline{(\overline{A} + C)}} = f$ (5)

5.3 $\overline{\overline{(AB + BC)}.\,\overline{B} + (A + B)} = f$ (5)

[15]

QUESTION 6

Design a circuit diagram that will supply a binary ONE at the output if the majority of the inputs (A, B and C) are equal to a binary ONE.

Make use of Boolean algebra and De Morgan's theorems to design the circuit diagram. The circuit diagram must make use of a minimum number of NAND gates.

The following must be indicated clearly:

- Truth table

[10]

- Application of Boolean algebra and De Morgan's theorems
- Circuit diagram with the minimum number of NAND gates

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QUESTION 7

7.1 Give a description of *fan out.* (4)

7.2 How many combinations are possible on a truth table with THREE variables? (1)

7.3 By making use of square-wave pulses, indicate the following on the pulses:

7.3.1 Mark period

7.3.2 Space period

7.3.3 Leading edge

7.3.4 Trailing edge

(4 × 1) (4)

7.4 Draw the symbol and the truth table of the full adder. (9)

7.5 Draw a block diagram for an 8 to 1 line multiplexer and discuss the purpose of the select lines in the block diagram. (7)

7.6 Draw a neat, labelled sketch of a 4-bit ring counter making use of D flip-flops. (5)
[30]

TOTAL SECTION B: 75
GRAND TOTAL: 100

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